# COMP396 Weekly Meeting Minutes

# (Week 1)

**Date & Time:** Thursday, September 25th, 2025, 14:00-17:00

**Location:** Sydney Jones Library / Abercromby Study Room 1

**Note Taker:** Zhao Saria

**Attendees:** Xue Yudan, Zhao Saria, Zhang Olivia, Wong Haw

## 1. Agenda:

- Self-introductions & Role Discussion
- Review of Project Objectives & Assignment Criteria
- Alignment on Technology Stack & Development Environment
- Analysis of "The Exercises" & Task Breakdown
- Future Meeting Planning

## 2. Discussion & Decisions:

### 2.1. Project Understanding & Consensus

- **Core Objective:** The primary goal is to collaboratively design, implement, test, and evaluate a robust automated financial trading strategy. The final deliverables emphasize not only the source code but also a comprehensive final report and a clear presentation.
- **Key Deliverables:** Identified four key assessment milestones
    - Assignment 1 (20%): Design Presentation and Report (Due: 13 Nov) - This is our immediate priority.

- Assignment 2 (30%): Evaluation of Trading Strategies (Due: 18 Mar 2026).
- Assignment 3a/3b (50%): Final Report, Source Code & Peer Review (Due: 29 Apr 2026).

### 2.2. Technology Stack & Development Environment

Based on the "Python Resources" and "The Exercises" sections provided on Canvas, we unanimously agree to standardize the following:

- **Language:** Python
- **Core Framework:** Backtrader
- **Libraries:** Pandas, NumPy, Matplotlib, etc.
- **IDE:** Visual Studio Code (VS Code)
- **Version Control:** We will use Git and a private GitHub repository to manage our codebase, ensuring effective collaboration and version control.
- **Communication:** WeChat will be used for daily communication and quick queries.

### 2.3. Task Planning & Strategy

All team members reviewed "The Exercises" provided in the course materials and agreed that these would serve as an excellent guide for the initial phase of the project.

- **Initial Workflow:**
  - **Environment Setup:** Prioritize the configuration of the IDE and Git repository.
  - **Example Analysis:** Conduct a deep dive into the provided example strategies (e.g., rsi_contrarian, bbands_contrarian) to build a foundational understanding.
  - **Technical Skill-Up:** Progressively work through the exercises to implement more advanced features, such as limit orders, expanded data stores (OHLCV), technical indicators, and position sizing.
- **Strategy Ideation:** While the immediate focus is on mastering the Backtrader framework, the team will begin sourcing ideas from the recommended blogs and podcasts for later discussion.

## 3. Task Allocation:

| ID | Module | Task Description | Owner | Due Date |
|---|---|---|---|---|
| A1 | Environment & Basics | Finalize the GitHub repository setup and complete local development environment configuration. | All Members | Sep. 30 |
| A2 | Core Concepts Research | Understand the rules for Budget Limit & Bankruptcy, the mechanics of Market vs. Limit Orders, and the PD Ratio calculation. | All Members | Oct. 2 |
| A3 | Example Strategy Analysis | Sub-group task:<br>- A3.1: Analyze the rsi_contrarian strategy.<br>- A3.2: Analyze the bbands_contrarian strategy. | Xue Yudan & Zhang Olivia (A3.1)<br>Wong Haw & Zhao Saria (A3.2) | Oct. 2 |
| A4 | Core Knowledge | Investigate how to implement Limit Orders in Backtrader. | All Members | Oct. 2 |
| A5 | Collaborative Report Structuring | Create a shared document in OneDrive. Set up the four main headings: Part 1: Individual Strategy Designs, Part 2: The Combined Strategy, Part 3: Breakdown of Teamwork, and Part 4: References. | Zhang Olivia | Oct. 2 |

## 4. Challenges & Issues

- **Unfamiliarity with Backtrader Framework:** Most of us realized that our knowledge of Backtrader (from COMP226) had faded. To address this, we agreed to dedicate the second week to a focused review of the Backtrader documentation and examples to re-familiarize ourselves with the framework.
- **Bridging the Gap from Theory to Practice:** While the course provides many technical tools and concepts, a key challenge will be integrating these individual components into a cohesive, logical, and effective trading strategy.
- **Time Constraints for Assignment 1:** The deadline of November 13th requires us to move quickly from learning the tools to designing and documenting our initial strategy. This demands efficient time management and clear

communication.

## 5. Supervisor Questions

- From the supervisor's experience, what are the most common mistakes or challenges that students encounter in this project, and how can we avoid them in advance?

## 6. Future Task List

Beyond the task allocation, we built a list based on the full list of "The Exercises" to guide our efforts leading up to Assignment 1.

- **Phase 2: Core Feature Implementation**
    - **Data Extension (Exercises #5, #6):** Expand data handling capabilities from close prices to full OHLCV (open, high, low, close, volume) data.
    - **Indicator Integration (Exercise #10):** Research and integrate technical indicators from libraries like ta or pandas-ta.
    - **Parameter Optimization (Exercise #9):** Learn and apply basic parameter optimization using Backtrader's optstrategy or tools like optuna.
- **Phase 3: Risk & Portfolio Management**
    - **Position Sizing (Exercise #11):** Research and implement fundamental position sizing techniques.
    - **Capital Management (Exercises #12, #13):** Simulate trading within the £1,000,000 budget and explore the implementation and risks of Shorting.
- **Phase 4: Analysis & Reporting**
    - **Performance Analytics (Exercises #8, #14):** Analyze advanced metrics such as the proportion of time in the market and correlation of returns.
    - **Automated Reporting (Exercises #15, #16):** Develop scripts (e.g., in a Jupyter Notebook) to generate comprehensive PDF evaluation reports.

## 7. Next Meeting

- **Date & Time:** Friday, October 3rd, 2025, 14:00-17:00
- **Location:** Sydney Jones Library / Abercromby Study Room 1
- **Key Agenda Items:**
    - Confirm completion of foundational tasks (A1-A5).
    - Brief presentations on our findings from the example strategies (A3).
    - Discuss the implementation of Limit Orders based on the research from task A4.
    - Propose initial ideas for the three individual strategies.
    - Define and assign tasks for Phase 2 ("Core Feature Implementation").